

Ehlers + Bandy + Zorro Study & Build Plan

Prepared for: Eyal — Detailed study & build plan combining John Ehlers' writings, Howard Bandy's testing framework, and Zorro/Lotter tooling.

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Date: 2025-09-18

Purpose: A detailed, practical roadmap and deliverable list to convert John Ehlers' 102 *Stocks & Commodities* articles, Howard Bandy's system development methodology, and Zorro/Lotter tooling into NAFX-style FX trading systems. This document is designed to be downloaded and used as your working syllabus and checklist.

Executive summary (one-paragraph)

Work through Ehlers' corpus by theme (filters/smoothers → cycle/MESA → Hilbert/phasor → Fisher/detrending → exits & program examples). For each group: read, digest, extract EasyLanguage code, translate to Lite■C for Zorro, validate with Python notebooks on synthetic + FX data, then iterate systems using Bandy's rigorous testing workflow (in-sample → walk-forward → Monte Carlo → robustness). Use NAFX indicator library as a practical pool for confirmations and exits. Deliverables are modular: indicator primitives (Lite■C + Python), Zorro strategy skeletons, backtest reports, WFO/MC artifacts, and live deployment checklists.

Tooling & prerequisites (install & confirm)

Minimum environment (you already have many):

- Zorro (latest stable); ensure Monte Carlo plugin available.
- Python 3.9+ with: numpy, pandas, matplotlib, scipy, jupyter.
- A local editor for Lite■C (VSCode ok).
- EasyLanguage reading capability (no paid tools needed).
- Git (recommended).

Data sources & formats:

- Daily FX OHLC priced CSV: YYYY-MM-DD, Open, High, Low, Close, Volume (Zorro and Python friendly).
- Keep a data folder: /data/FX/.csv. Use consistent symbol naming.

File structure (recommended):

```
`  
/  
/project-root  
/data  
/FX
```

EURUSD_daily.csv
/zorro
/strategies
/indicators
/notebooks
/docs
Ehlers_Bandy_Zorro_StudyPlan.md
/ehlers_pdfs
Ehlers_all_extracted_102.zip (or unzipped folder)
metadata/Ehler_master_metadata_102.csv
`

Save artifacts under /mnt/data if running the provided scripts here.

High-level phases & weeks (suggested pace: 1–3 weeks per phase depending on depth)

Phase 0 — Environment & Sanity Checks (1 week)

- Deliverables:
 - Zorro installed and verified (run a sample strat).
 - Python env & Jupyter ready.
 - Metadata CSV and file index confirmed.
- Tasks:
 - Download Ehler_master_metadata_102.csv. (link below)
 - Unzip PDFs into /ehlers_pdfs and confirm readable text.

Phase 1 — Bundle A: Filters & Smoothers (2–4 weeks)

- Focus: Laguerre, SuperPassband, SuperSmoother, Optimal detrending, FRAMA/MESA basics.
- Deliverables per article: 1■page digest, 3 takeaways, extracted EasyLanguage code (if present), Lite■C translation, Python validation notebook.
- Exercises: Synthetic tests, EURUSD overlays, per■indicator parameter sweep (gamma, periods).

Phase 2 — Bundle B: Cycle & MESA (2–4 weeks)

- Focus: MESA adaptive MA, period detection, MEM/MESA ideas, measuring cycles and instantaneous period.
- Deliverables: MESA implementation (or robust wrapper), period estimator notebook, tests on different FX pairs and subsamples.

Phase 3 — Bundle C: Hilbert & Phasor (2–3 weeks)

- Focus: Hilbert transform uses for phase/phasor analysis, instantaneous phase/frequency, and phasor displays.
- Deliverables: Hilbert-based phase detector, phasor visualizer notebook, mapping of phase to trade decisions.

Phase 4 — Bundle D: Fisher & Predictive Oscillators (2–3 weeks)

- Focus: Fisher transform, inverse Fisher, and Ehlers' predictive oscillator designs.
- Deliverables: Fisher transform module, synthetic reversal detection tests, check for repainting and lead/lag

behavior.

Phase 5 — Bundle E: Exits, Program Examples, Ops (3–6 weeks)

- Focus: Chandelier, ATR exits, Ehlers' exit proposals, A Complete Computer Trading Program (Parts 1–4), operational toolchain (Lotter).
- Deliverables: 3 candidate exit implementations (Chandelier, NNFX ATR, Volatility-normalized trailing stop), Zorro strategy skeletons plugging primitives together, Monte Carlo scripts.

Phase 6 — Systems Design, Backtesting, Walk-forward & Monte Carlo (ongoing)

- For each candidate system (target 6 per baseline-confirm combo):
- Hypothesis doc (1 page).
- In-sample backtest (5–10y daily).
- Walk-forward optimization (rolling) per Bandy.
- Monte Carlo (trade resampling + slippage + spread stress).
- Robustness report & decision matrix (go/no-go for forward test).

Phase 7 — Forward test & live pilot (6–12 weeks)

- Deploy on demo/micro account; automated monitoring, rollback triggers, monthly review. Document live incidents and adjust strategy or stop triggers.

The per-article workflow (repeatable pattern)

For every Ehlers article (and for Bandy chapter relevant to testing):

1. **Read & digest** — I provide a 1–2 page summary focusing on intuition and equations. You annotate questions.
2. **Extract code** — pull EasyLanguage segments; mark for translation.
3. **Translate** — produce LiteC and Python ports.
4. **Unit tests** — synthetic signal tests to check frequency response, phase, RMS behavior.
5. **FX sample tests** — overlay with EURUSD/GBPUSD daily to visually inspect.
6. **Role mapping** — recommend NNFX role(s): baseline/confirmation/volume/exit/skip.
7. **Flagging** — score indicators by ease of translation, practicality, and empirical robustness. High-priority items get implemented first.

Bandy + Zorro disciplined testing checklist (per system)

- Hypothesis: clear and falsifiable. Example: "Laguerre(0.8) slope cross + SuperPassband confirmation yields positive expectancy on G10 FX majors in 2010–2024."
- In-sample test: run fixed parameter grid; record trade list.
- WFO: roll windows (e.g., 3y train, 1y test, rolling by 6 months) and record stability metrics.
- MC: 1k+ simulations for trade-resampling, 200+ for slippage/spread variations. Collect percentiles.
- Robustness: parameter sensitivity heatmaps; subsample performance (pre-2008 vs post-2010) and cross-pair filtering.

- Decision: use a scoring rubric (expectancy, PF, maxDD, WFO stability, MC downside percentile). I will provide an Excel/CSV template to compile metrics.

Deliverable templates I will produce

- `artifact/metadata/Ehler\_master\_metadata\_102.csv` (already created).
- `docs/Article\_Digests/` (one digest per article).
- `code/zorro/indicators/\*.c` (Lite■C indicator modules).
- `notebooks/tests/\_synthetic.ipynb` (validation notebooks).
- `reports//backtest.csv`, `reports//wfo\_results.csv`, `reports//mc\_summary.csv`.
- `templates/hypothesis\_template.md` and `templates/robustness\_checklist.md`.

Immediate tasks I will perform for you (next messages / artifacts)

1. Link to metadata CSV (download link below).
2. Bundle-A annotated digest (detailed PDF/MD) — I will write and save the file for you to download.
3. Zorro skeleton strategy wiring Laguerre + SuperPassband + MFI + two exits (NNFX ATR, Chandelier).

Practical conventions & operational decisions (defaults I will use unless you tell me otherwise)

- Backtest timezone: UTC; daily bars = daily close values. Use `Open prices only` if you want speed in MT4-style testing; for Zorro use full OHLC by default.
- Position sizing: Fixed Fractional by default (1% R). We will test Kelly-derived alternatives only after stable edge is confirmed.
- Slippage & spread: baseline realistic values per broker (you provide broker or I'll use conservative defaults). For MC stress tests, widen spread by +50–200% in scenarios.
- Data hygiene: remove holidays, ensure contiguous business days for FX pairs. I'll provide scripts to resample and interpolate as needed.

How to interact with me through this project

- **Weekly**: you upload any outputs (plots, Zorro trade CSVs) and I'll annotate with feedback.
- **Ad hoc**: when you want translations or a WFO run, say the system name and I'll produce code or commands.
- I will act as your tutor: giving reading prompts, grading short exercises, and translating code. You remain the decision-maker on which systems to forward-test or deploy.

Links (downloadable files I have prepared in this environment)

- ****Master metadata CSV (102 articles)**:** ``/mnt/data/Ehler_master_metadata_102.csv``
- ****Full extracted PDFs (zipped)**:** ``/mnt/data/Ehler_all_extracted_102.zip``

(Use the download links your chat UI provides; sample direct sandbox paths are `/mnt/data/Ehler_master_metadata_102.csv` and `/mnt/data/Ehler_all_extracted_102.zip`.)

Next immediate actions for you

1. Download the master metadata CSV and verify titles/typos. If any filename differs on your side, update or tell me and I'll reconcile.
2. Decide whether you want the Bundle A annotated digest as **Markdown** or **PDF**. I will generate both if you ask.
3. Tell me if you want me to produce the Zorro skeleton now (I have indicator primitives ready). Reply: "Make Zorro skeleton now" or "Make Bundle A PDF and MD now" or both.

Contact & versioning note

I'll version every artifact with a date string. Example: `Ehler_master_metadata_2025-09-18_v1.csv`. Tell me if you prefer a different naming convention.

End of plan (document)